

Batch:	Roll No.:
Experiment / assignment / tutorial No. 09	
Grade: AA / AB / BB / BC / CC / CD / DD	
Signature of the Staff In-charge with date	

Title: Time Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM) using MATLAB

Objectives:

1. To understand the concept of TDM and FDM and write a MATLAB program for it.

OUTCOME:

1. understand working of analog communication transmitter and receiver systems
2. compare Pulse modulation systems and multiplexing techniques.
3. understand noise effect on the performance of communication systems.

- **Time Division Multiplexing (Write one paragraph of TDM concept along with diagram)**
- **MATLAB Program:**
 - **Generate two signals with different frequency**
 - **Multiplex them in time domain to demonstrate TDM concept**
 - **Plot both signal and multiplexed signal**
- **Frequency Division Multiplexing (Write one paragraph of FDM concept along with diagram)**
- **MATLAB Program:**
 - **Generate two signals with different frequency**
 - **Multiplex them in time domain to demonstrate FDM concept**
 - **Plot both signal and multiplexed signal**
- **Post lab activity:**
 - Modify the code of TDM for multiplexing three signals and copy the code with output. (2 marks)
 - Modify the code of FDM for multiplexing three signals and copy the code with output. (2 marks)
 - Write applications for FDM and TDM (1 mark)

Signature of faculty in-charge